

SUMMARY

PhD candidate researching **timing integrity and predictability of real-time operating systems in high-performance computing**. Pursuing a career as a **systems and performance engineer** — with hands-on expertise in **compiler optimization, GPU scheduling/optimization, parallel computing, Linux-based networks and real-time safety**, backed by publications and production deployments.

PUBLICATIONS

- **OpenMP-RT-Offload**: Real-Time Support for OpenMP GPU Offloading — Under Submission
- **OpenMP-Q**: Quantum Task Offloading Extension for OpenMP — IWOMP 2025
- **T-Tex**: Timed Security in Multi-Threaded Real-Time Systems — ICCPS 2025
- **T-Pack**: Timed Network Security for Real-Time Systems — ISORC 2021

EDUCATION

PhD, Computer Science <i>Compiler Optimization, Embedded/Real-Time Systems, Parallel Systems</i>	North Carolina State University, Raleigh, NC <i>Aug 2020 – Present</i>
MS, Computer Science <i>Operating Systems, Linux Networking, Internet Protocol, Graph Theory</i>	North Carolina State University, Raleigh, NC <i>Aug 2018 – Apr 2020</i>
B.Tech., Computer Science <i>Parallel and Concurrent Systems, Cloud Computing, Agent Based Intelligent Systems</i>	Vellore Institute of Technology, India <i>Jul 2014 – Apr 2018</i>

WORK EXPERIENCE

AIM Intelligent Machines <i>Performance Optimization Engineer</i>	Monroe, WA <i>May 2024 – Aug 2024</i>
• Reduced CPU process latency by $\geq 80\%$ in a real-time autonomous robotics pipeline (Autonomous Dozer) through targeted profiling (Perf, Nsight Compute, isolated execution analysis). • Implemented MPS-integrated GPU scheduling framework, cutting GPU latency step time by $\approx 9\%$ in the pipeline.	
Programming Systems, Uber <i>Compiler Optimization Engineer</i>	San Francisco, CA <i>May 2022 – Aug 2022</i>
• Implemented profile-guided code layout optimization (PGO) in the Go linker using a cross-package weighted call graph with C3 heuristics • Optimized function layout for cache utilization across Google-scale multi-package Go benchmarks	

RESEARCH EXPERIENCE

Graduate Research Assistant	North Carolina State University, Raleigh, NC <i>Aug 2019 – Present</i>
------------------------------------	---

Research Advisor: Dr. Frank Mueller

OpenMP-Q: Quantum Task Offloading for HPC (IWOMP 2025):

- Designed an OpenMP language extension integrating quantum task offloading into HPC workflows via POSIX pipes, achieving up to **2.9 $\times$ speedup**
- Built a quantum-circuit library supporting reverse offloading and broad hardware compatibility

OpenMP-RT-Offload: Real-Time GPU Scheduling for OpenMP (Under Submission):

- Designed an application-transparent interposer using NVIDIA Green Contexts and per-partition priority-inheritance queues to deliver real-time scheduling for OpenMP GPU offloading without modifying application code or the OpenMP runtime
- Reduced worst-case response time by **12–25ms** for high-priority tasks vs. baseline
- For 8 tasks over 2 partitions: **59%** miss rate reduction vs. baseline, **50%** vs. state-of-the-art, and **zero misses** for the second-highest-priority task

T-TeX: Timed Security in Multi-Threaded Real-Time Systems (ICCPs 2025):

- Developed OpenMP code region identification using LLVM/Clang to detect and prevent timing-based delay attacks in real-time systems
- Designed a kernel-to-runtime communication interface to monitor context switches and secure timers, enabling worst-case execution time (WCET) evaluation
- Achieved **100% attack detection** for 60 μ s delay attacks with minimal runtime overhead on an autonomous driving benchmark

T-Pack: Network Security for Safety Critical Real-Time Systems (ISORC 2021):

- Built a Linux kernel module using socket buffers for real-time transmission timing analysis, detecting **95–100% of attacks** with ≈ 0.012 ms overhead
- Extended to all major network protocols with IPSec encryption compatibility

Advisor: Dr. Hung-Wei Tseng

In-Network Processing Analysis:

- Developed Apache Spark/Hadoop benchmarks to identify network congestion bottlenecks in SDN/5G edge computing environments

TEACHING EXPERIENCE

Graduate Teaching Assistant — Parallel Systems, NC State University

- Mentored graduate students on CUDA/NVIDIA GPU programming, MPI cluster programming, OpenMP/OpenACC multithreading, and system performance/power optimization

PROJECTS

Autonomous Vehicle Simulator (Carla):

- Designed a PID controller and CNN model to simulate autonomous driving; **100% of 10 concurrent vehicles** successfully followed traffic signs

Cloud Provider with DNS-as-a-Service:

- Built a virtual private cloud using Linux LXD containers with hierarchical DNS (bind9) and load balancing as default services

Peer-to-Peer File Sharing System:

- Implemented a dynamic TCP/UDP file sharing system with stop-and-wait reliability for point-to-multipoint transfer, achieving **95% network efficiency**

SKILLS

Languages: C, C++, Python, Go

High Performance Computing: OpenMP, MPI, CUDA, OpenACC

Compilers & Optimization: LLVM/Clang, DynamoRIO, Go Compiler, PGO, Profile-Guided Optimization

Systems Programming: Linux Kernel Modules, Network Stack, Process/GPU Scheduling, Signals, Timers, Real-Time Inference, Safety-Critical Systems

ML: CNN, PID Controller, Carla Simulator, Autonomous Driving Pipeline

Distributed & Networked Systems: Apache Spark, Hadoop, SDN, IPSec, KVM, CloudStack